N_2 of Group I, Row III: UV logarithms, finite terms and the longitudinal logarithms, step by step

1 Starting expression

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\\
&\times\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}\right.\\
&-\frac{p^+}{(k^+-p^+)}\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&-\frac{(k^+-p^+)}{p^+}\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]\\
&\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\\
&\times\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(y)+N_c\,U^{af}(x)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
\tag{1.1}
$$

2 Step 0: shorthand

$$\mathcal{N}\equiv\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+},\qquad\frac{\mathcal{N}}{2\pi}=\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+},\tag{2.1}$$

$$\xi\equiv\frac{p^+}{k^+},\qquad\lambda\equiv\frac{\Lambda}{k^+},\tag{2.2}$$

$$\Phi(p^+)=e^{-ik\cdot(z-x)\frac{p^+}{k^+}},\qquad D(p^+)=p^+(y-x)^2+(k^+-p^+)(y-z)^2,\tag{2.3}$$

$$T^{im}=\frac{(x'-w')^i(z-x)^m}{(x'-w')^2(z-x)^2},\tag{2.4}$$

$$B^{im}=\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2},\tag{2.5}$$

$$A^{im}=\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2},\tag{2.6}$$

and $R^{im}(\xi)$ denotes the four non-singular brackets of (1.1) (the $d_\perp$, $(k^+-p^+)\delta$, $p^+\delta$ and last brackets). In this notation the bracket of (1.1) is

$$R^{im}(\xi)-\frac{\xi}{1-\xi}A^{im}-\frac{1-\xi}{\xi}B^{im}.\tag{2.7}$$

Split the colour factor into the N_{2a} and N_{2b} parts:

$$\mathcal{C}^{(a)}=\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[-f^{adc}f^{fbe}U^{db}(x)U^{ce}(z)+N_cU^{af}(x)\Big]\rho^{b'}(x')\rho^f(y),\tag{2.8}$$

$$\mathcal{C}^{(b)} = \Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_c U^{af}(y) - N_c U^{af}(x)\Big]\rho^{b'}(x')\rho^{f}(y), \tag{2.9}$$

$$\mathcal{C}^{(b)}_z = \Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_c U^{af}(y) - N_c U^{af}(z)\Big]\rho^{b'}(x')\rho^{f}(y), \tag{2.10}$$

$$\mathcal{C}_2 = \mathcal{C}^{(a)} + \mathcal{C}^{(b)} = \Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[- f^{adc}f^{fbe}U^{db}(x)U^{ce}(z) + N_c U^{af}(y)\Big]\rho^{b'}(x')\rho^{f}(y), \tag{2.11}$$

and define

$$K_S = \frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2}. \tag{2.12}$$

3 Step 1: where the x integral can diverge

Fix $0 < p^+ < k^+$; the only variable that is integrated over and appears in the kernel singularly is x. Check the three regions.

- **Large x.** $T \sim 1/|x|$, $D \sim p^+x^2$, and all brackets are at most $O(1)$, so the integrand falls at least like $1/|x|^3$. This is finite.
- **$x \to y$.** The brackets have at most $1/|y-x|$, which is integrable in two dimensions. At $p^+ \to k^+$ one has $D \to k^+(y-x)^2$, but $\int_0^{k^+} dp^+/D = \frac{\log[(y-z)^2/(y-x)^2]}{(y-z)^2-(y-x)^2}$, which is only a logarithm and is integrable. This is finite.
- **$x \to z$.** Several terms go as $1/(x-z)^2$, which is logarithmically divergent in two dimensions. This is the UV region; Step 2 works it out.

4 Step 2: the integrand as $x \to z$

Put $r = x - z \to 0$, $n = y - z$, $a = x' - w'$, so $y - x = n - r$ and $(y-x)^2 - (y-z)^2 = -2\,n\cdot r + r^2$. At $r = 0$: $\Phi = 1$ and $D = k^+n^2$. Keep only the $1/r^2$ terms and average over the direction of r with $\langle r^i r^j\rangle = \frac{r^2}{2}\delta^{ij}$ (in d dimensions $\frac{r^2}{d}\delta^{ij}$ together with $d_\perp = d$, which gives the same results).

$$d_\perp \text{ term:}\quad \frac{d_\perp}{2}\,\xi(1-\xi)\,\frac{(-a\cdot r)(-2\,n\cdot r)}{a^2r^4} \;\to\; \xi(1-\xi)\,\frac{a\cdot n}{a^2r^2}, \tag{4.1}$$

$$(1-\xi)\delta \text{ term:}\quad \text{bracket} = \frac{2\,n\cdot r}{r^2} + O(1) \;\Rightarrow\; (1-\xi)\frac{(-a\cdot r)(2\,n\cdot r)}{a^2r^4} \;\to\; -(1-\xi)\frac{a\cdot n}{a^2r^2}, \tag{4.2}$$

$$-\tfrac{1-\xi}{\xi}B \text{ term:}\quad T^{im}B^{im} = -\frac{a\cdot n}{a^2r^2} + O(1/r) \;\Rightarrow\; +\frac{1-\xi}{\xi}\frac{a\cdot n}{a^2r^2}, \tag{4.3}$$

$$\xi\delta \text{ term:}\quad \text{bracket} = \frac{2\,n\cdot r}{r^2} + O(1) \;\Rightarrow\; -\xi\,\frac{a\cdot n}{a^2r^2}, \tag{4.4}$$

$$\text{last bracket:}\quad -\frac{2\,n^m r^i}{r^2} + O(1) \;\Rightarrow\; \frac{a^i(-r^m)}{a^2r^2}\Big(-\frac{2n^m r^i}{r^2}\Big) \to +\frac{a\cdot n}{a^2r^2}, \tag{4.5}$$

$$-\tfrac{\xi}{1-\xi}A \text{ term:}\quad T^{im}A^{im} = -\frac{a\cdot n}{a^2r^2} + O(1/r) \;\Rightarrow\; +\frac{\xi}{1-\xi}\frac{a\cdot n}{a^2r^2}. \tag{4.6}$$

Adding these up, the constant terms cancel ($-(1-\xi) - \xi + 1 = 0$), and with $1/D \to 1/(k^+n^2)$:

$$\text{integrand of (1.1)} \xrightarrow{x\to z} \frac{1}{k^+}\,P(\xi)\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2\,(x-z)^2} \times (\text{colour at } x=z), \qquad P(\xi) = \xi(1-\xi) + \frac{\xi}{1-\xi} + \frac{1-\xi}{\xi}. \tag{4.7}$$

This was also checked numerically: the angle-averaged integrand divided by K_S tends to $P(\xi)$ (for example 4.41, 2.25 and 9.2011 at $\xi = 0.2,\ 0.5,\ 0.9$), and the difference falls like r^2.

5 Step 3: colour at $x \to z$, and why only N_{2b} has a UV divergence

Adjoint Wilson lines leave f invariant, so $U^{db}(z)U^{ce}(z)f^{fbe} = f^{dcg}U^{gf}(z)$. Then

$$-f^{adc}f^{fbe}U^{db}(z)U^{ce}(z) + N_c U^{af}(z) = -f^{adc}f^{dcg}U^{gf}(z) + N_c U^{af}(z) = -N_c U^{af}(z) + N_c U^{af}(z) = 0. \tag{5.1}$$

This was checked numerically for a random SU(3) matrix: the residual is 2×10^{-15}.

- N_{2a} (colour $\mathcal{C}^{(a)}$): the colour vanishes linearly at $x \to z$, so the integrand is at most $O(1/r)$. N_{2a} is **UV finite**.
- N_{2b} (colour $\mathcal{C}^{(b)}$): $\mathcal{C}^{(b)} \to \mathcal{C}^{(b)}_z \neq 0$, so N_{2b} is **UV divergent**, with the coefficient given by (4.7).

6 Step 4: subtraction

Define the subtraction term, which has exactly the $x \to z$ behaviour (4.7):

$$S(p^+) = \frac{1}{k^+} P(\xi)\, K_S\, \mathcal{C}_z^{(b)}. \tag{6.1}$$

At $x \to z$ we have $\frac{(y-x)\cdot(y-z)}{(y-x)^2} \to 1$, so $K_S \to \frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(x-z)^2}$. In addition K_S is integrable at $x \to y$ and at large x (where it falls like $|x|^{-3}$), so S adds no new divergence. Write, identically,

$$N_2 = \underbrace{\mathcal{N}\int \frac{dp^+}{2\pi}\int e^{-ik\cdot(w'-z)}\, S}_{N_{2b\,\mathrm{UV}}} + \underbrace{\mathcal{N}\int \frac{dp^+}{2\pi}\int e^{-ik\cdot(w'-z)}\Big[\text{integrand of (1.1)} - S\Big]}_{N_{2\,\text{finite}}}. \tag{6.2}$$

The second term is UV finite, so we set $d = 2$ ($d_\perp = 2$) in it. All the UV divergence is in the first term.

7 Step 5: UV part

5a. The x integral in two dimensions (to see the divergence)

Put $x = z + t$, $s = |t|$, $n = y - z$. Writing $n^2 - ns\cos\varphi = \frac{1}{2}\big[(n^2 + s^2 - 2ns\cos\varphi) + (n^2 - s^2)\big]$ and using $\int_0^{2\pi} \frac{d\varphi}{A - B\cos\varphi} = \frac{2\pi}{\sqrt{A^2 - B^2}}$:

$$\int_0^{2\pi} d\varphi\, \frac{n^2 - ns\cos\varphi}{n^2 + s^2 - 2ns\cos\varphi} = \pi + \pi\,\frac{n^2 - s^2}{|n^2 - s^2|} = \begin{cases} 2\pi, & s < |n| \\ 0, & s > |n| \end{cases} \tag{7.1}$$

(checked numerically). Hence

$$\int_x \frac{(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2} = 2\pi \int_0^{|y-z|} \frac{ds}{s}, \tag{7.2}$$

which is logarithmically divergent at $s \to 0$, that is at $x \to z$ (UV). It is finite at large x and at $x \to y$.

5b. The same integral in $d = 2 - \epsilon$

Using a Feynman parameter, $\frac{1}{(y-x)^2(x-z)^2} = \int_0^1 \frac{d\alpha}{[(t-\alpha n)^2 + \alpha(1-\alpha)n^2]^2}$, and shifting $t \to t + \alpha n$:

$$\begin{aligned} \mu^{-\epsilon}\int d^d t\, \frac{(n-t)\cdot n}{(n-t)^2 t^2} &= \mu^{-\epsilon}\, n^2\, \pi^{d/2}\Gamma\big(2 - \tfrac{d}{2}\big) \int_0^1 d\alpha\, (1-\alpha)\big[\alpha(1-\alpha)n^2\big]^{\frac{d}{2}-2} \\ &= \pi^{d/2}\,\Gamma\big(2 - \tfrac{d}{2}\big)\,\frac{\Gamma(\frac{d}{2}-1)\Gamma(\frac{d}{2})}{\Gamma(d-1)}\,(\mu^2 n^2)^{-\epsilon/2} \\ &= \pi\left[\frac{2}{\epsilon_{\mathrm{UV}}} + 2\gamma_E + \log\frac{(y-z)^2\mu_{\overline{MS}}^2}{4}\right] + O(\epsilon), \end{aligned} \tag{7.3}$$

with $\mu_{\overline{MS}}^2 = 4\pi e^{-\gamma_E}\mu^2$ and $\epsilon_{\mathrm{UV}} = -\epsilon$; this was checked with sympy. This is the same convention as the Divergent file.

5c. The p^+ integral, done exactly

$K_S \mathcal{C}_z^{(b)}$ does not depend on p^+, so

$$\begin{aligned} \frac{1}{k^+}\int_\Lambda^{k^+-\Lambda} dp^+ P(\xi) &= \int_\lambda^{1-\lambda} d\xi\Big[\xi(1-\xi) + \Big(\frac{1}{1-\xi} - 1\Big) + \Big(\frac{1}{\xi} - 1\Big)\Big] \\ &= \Big(\frac{1}{6} - \lambda^2 + \frac{2\lambda^3}{3}\Big) + 2\log\frac{1-\lambda}{\lambda} - 2(1-2\lambda) = 2\log\frac{k^+-\Lambda}{\Lambda} - \frac{11}{6} + \frac{4\Lambda}{k^+} - \frac{\Lambda^2}{k^{+2}} + \frac{2\Lambda^3}{3k^{+3}}. \end{aligned} \tag{7.4}$$

5d. Result for the UV part

The prefactor is $\frac{\mathcal{N}}{2\pi}\cdot\pi\cdot N_c = \frac{1}{(2\pi)^3}\frac{g^4 N_c}{16\pi^4}\frac{1}{k^+}$, so

$$\begin{aligned} \frac{d^3 N_{2b\,\mathrm{UV\,NLO}}}{d^3 k} &= \frac{1}{(2\pi)^3}\,\frac{g^4 N_c}{16\pi^4}\,\frac{1}{k^+}\left[2\log\Big(\frac{k^+-\Lambda}{\Lambda}\Big) - \frac{11}{6} + \frac{4\Lambda}{k^+} - \frac{\Lambda^2}{k^{+2}} + \frac{2\Lambda^3}{3k^{+3}}\right] \\ &\quad \times \int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-z)^2\mu_{\overline{MS}}^2}{4}\right] \end{aligned}$$

$$\times\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[U^{af}(y)-U^{af}(z)\Big]\rho^{b'}(x')\,\rho^{f}(y) \tag{7.5}$$

$$\xrightarrow{\Lambda\to 0}\ \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\left[2\log\frac{k^+}{\Lambda}-\frac{11}{6}\right]\int_{y,z,x',w'}(\cdots). \tag{7.6}$$

Here $2\log\frac{k^+}{\Lambda}=2\log\frac{\vee}{\Lambda}-2\log\frac{\vee}{k^+}$. The $-\frac{11}{6}\times$UV piece is the one that combines with Row II ($\frac{11}{3}$) in the Divergent file. The blue piece is a longitudinal log multiplying the UV bracket; Step 7 shows that it cancels against the $-2\log$ term of the finite part when the x integral is undone.

8 Step 6: finite part (before the plus prescription)

$$\frac{d^3N_{2\,\text{finite}}}{d^3k}=\mathcal{N}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\left\{\frac{\Phi}{D}\,T^{im}\Big[R^{im}(\xi)-\frac{\xi}{1-\xi}A^{im}-\frac{1-\xi}{\xi}B^{im}\Big]\mathcal{C}_2-\frac{1}{k^+}P(\xi)\,K_S\,\mathcal{C}_z^{(b)}\right\}, \tag{8.1}$$

with $d_\perp=2$. This is UV finite at every p^+ by Steps 2 to 4, but it still has the Λ cut-offs. The singular coefficients are $\frac{1-\xi}{\xi}$ (at $p^+\to 0$) and $\frac{\xi}{1-\xi}$ (at $p^+\to k^+$), together with the same two inside $P(\xi)$.

9 Step 7: plus prescription and the longitudinal logarithms

7a. Definitions

$$\Big[\frac{k^+}{p^+}\Big]_+ f(p^+)\equiv\frac{k^+}{p^+}\big[f(p^+)-f(0)\big],\qquad \Big[\frac{k^+}{q^+}\Big]_+ h(q^+)\equiv\frac{k^+}{q^+}\big[h(q^+)-h(0)\big],$$

$$\Big[\frac{k^+}{k^+-p^+}\Big]_+ f(p^+)\equiv\frac{k^+}{k^+-p^+}\big[f(p^+)-f(k^+)\big].$$

Then, exactly,

$$\int_\Lambda^{k^+-\Lambda}dp^+\frac{k^+}{p^+}f=\int_\Lambda^{k^+-\Lambda}dp^+\Big[\frac{k^+}{p^+}\Big]_+ f+k^+f(0)\log\frac{k^+-\Lambda}{\Lambda}\ \xrightarrow{\Lambda\to 0}\ \int_0^{k^+}dp^+\Big[\frac{k^+}{p^+}\Big]_+ f+k^+f(0)\log\frac{k^+-\Lambda}{\Lambda}. \tag{9.1}$$

7b. The $p^+\to 0$ end of the main term

Write $-\frac{1-\xi}{\xi}B=-\frac{k^+}{p^+}B+B$ and set $f(p^+)=\Phi\,T^{im}B^{im}/D$. At $p^+=0$: $\Phi=1$ and $D=k^+(y-z)^2$, so

$$T^{im}B^{im}=\frac{(x'-w')\cdot(y-z)\,(x-z)\cdot(z-x)}{(x'-w')^2(z-x)^2(x-z)^2}+\frac{(x'-w')\cdot(y-z)\,(y-x)\cdot(z-x)}{2(x'-w')^2(z-x)^2(y-x)^2}, \tag{9.2}$$

$$-k^+f(0)=+\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\left[\frac{1}{(x-z)^2}+\frac{(x-z)\cdot(y-x)}{2(x-z)^2(y-x)^2}\right]. \tag{9.3}$$

7c. The $p^+\to k^+$ end of the main term (in $q^+=k^+-p^+$)

With $p^+=k^+-q^+$ and $\int_\Lambda^{k^+-\Lambda}dp^+=\int_\Lambda^{k^+-\Lambda}dq^+=-\int_{k^+-\Lambda}^{\Lambda}dq^+$, the term $-\frac{\xi}{1-\xi}A=-\big(\frac{k^+}{q^+}-1\big)A$ becomes

$$\int_{k^+-\Lambda}^{\Lambda}dq^+\Big(\frac{k^+}{q^+}-1\Big)h(q^+)=\int_{k^+}^{0}dq^+\Big\{\Big[\frac{k^+}{q^+}\Big]_+-1\Big\}h(q^+)-k^+h(0)\log\frac{k^+-\Lambda}{\Lambda}, \tag{9.4}$$

where $h(q^+)=e^{-ik\cdot(z-x)\frac{k^+-q^+}{k^+}}\,T^{im}A^{im}/\big[(k^+-q^+)(y-x)^2+q^+(y-z)^2\big]$. At $q^+=0$: $\widetilde{D}=k^+(y-x)^2$ and $e^{-ik\cdot(w'-z)}e^{-ik\cdot(z-x)}=e^{-ik\cdot(w'-x)}$, so

$$T^{im}A^{im}=\frac{(x'-w')\cdot(y-x)\,(z-x)\cdot(x-z)}{(x'-w')^2(z-x)^2(x-z)^2}-\frac{(x'-w')\cdot(y-x)\,(z-x)\cdot(y-z)}{2(x'-w')^2(z-x)^2(y-z)^2}, \tag{9.5}$$

$$-k^+h(0)=+e^{-ik\cdot(z-x)}\frac{(x'-w')\cdot(y-x)}{(x'-w')^2(y-x)^2}\left[\frac{1}{(x-z)^2}-\frac{(x-z)\cdot(y-z)}{2(x-z)^2(y-z)^2}\right]. \tag{9.6}$$

The endpoint coefficients of 7b and 7c were checked numerically: for random points and $\Lambda/k^+=10^{-7}$, the original $\int_\Lambda^{k^+-\Lambda}$ agrees with the plus form plus the log terms to 10^{-8}, i.e. to $O(\Lambda/k^+)$.

7d. The subtraction term

Write $P(\xi)=\xi(1-\xi)+\frac{k^+}{k^+-p^+}-1+\frac{k^+}{p^+}-1$. Since $K_S\mathcal{C}_z^{(b)}$ is p^+ independent, each endpoint gives $K_S\mathcal{C}_z^{(b)}\log\frac{k^+-\Lambda}{\Lambda}$:

$$-\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}dp^+P(\xi)\,K_S\mathcal{C}_z^{(b)}=-\frac{1}{k^+}\int_0^{k^+}dp^+\Big[\xi(1-\xi)+\Big[\frac{k^+}{k^+-p^+}\Big]_+\!+\Big[\frac{k^+}{p^+}\Big]_+\!-2\Big]K_S\mathcal{C}_z^{(b)}-2\log\frac{k^+-\Lambda}{\Lambda}\,K_S\mathcal{C}_z^{(b)}. \quad (9.7)$$

The first term is left unevaluated, as decided. Evaluating it would give $+\frac{11}{6}K_S\mathcal{C}_z^{(b)}$, because $\int[\cdot]_+\times\text{const}=0$.

10 Step 8: result

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k}=\frac{d^3N_{2b\,\mathrm{UV}}}{d^3k}\ (\text{Step 5d})+\frac{d^3N_{2\,\mathrm{finite}}}{d^3k}\ (8a)+\frac{d^3N_{2\,\log}}{d^3k}\ (8b)+O(\Lambda/k^+). \quad (10.1)$$

8a. Finite part (Λ independent; $\mathcal{C}_2=\mathcal{C}^{(a)}+\mathcal{C}^{(b)}$)

$$\begin{aligned}
\frac{d^3N_{2\,\mathrm{finite\,NLO}}}{d^3k}=&\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\Bigg\{e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\\
&\times\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,[(y-x)^2-(y-z)^2]\,\frac{p^+(k^+-p^+)}{k^{+2}}\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\Big[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\Big]\\
&-\Big[\frac{k^+}{p^+}\Big]_+\Big[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\Big]+\Big[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\Big]\\
&+\frac{p^+\,\delta_{im}}{k^+}\Big[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\Big]\\
&-\Big[\frac{(y-x)^m(x-z)^i}{(x-z)^2}-\frac{(y-x)^m(y-z)^i}{2(y-z)^2}+\frac{(y-z)^m(x-z)^i}{(x-z)^2}+\frac{(y-x)^i(y-z)^m}{2(y-x)^2}\Big]\Bigg]\\
&\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[-f^{adc}f^{fbe}U^{db}(x)U^{ce}(z)+N_c\,U^{af}(y)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{k^+}\Big[\frac{p^+(k^+-p^+)}{k^{+2}}+\Big[\frac{k^+}{k^+-p^+}\Big]_+\!+\Big[\frac{k^+}{p^+}\Big]_+\!-2\Big]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}\\
&\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y)\Bigg\}\\
&+\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\int_{k^+}^0\frac{dq^+}{2\pi}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\times\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\Big[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\Big]\Big\{\Big[\frac{k^+}{q^+}\Big]_+-1\Big\}\\
&\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[-f^{adc}f^{fbe}U^{db}(x)U^{ce}(z)+N_c\,U^{af}(y)\Big]\rho^{b'}(x')\,\rho^f(y), \quad (10.2)
\end{aligned}$$

with $d_\perp=2$. To get the N_{2a} and N_{2b} pieces separately, replace the colour factor $\big[-ffUU+N_cU(y)\big]$ by $\big[-ffUU+N_cU(x)\big]$ for N_{2a}, and by $\big[N_cU(y)-N_cU(x)\big]$ for N_{2b}. The subtraction line belongs to N_{2b} only.

8b. Longitudinal logarithms (these go to the JIMWLK file)

$$\begin{aligned}
\frac{d^3N_{2\,\log}}{d^3k}=&\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\log\Big[\frac{k^+-\Lambda}{\Lambda}\Big]\int_{x,y,z,x',w'}\Bigg\{e^{-ik\cdot(w'-z)}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\Big[\frac{1}{(x-z)^2}+\frac{(x-z)\cdot(y-x)}{2(x-z)^2(y-x)^2}\Big]\mathcal{C}_2\\
&+e^{-ik\cdot(w'-x)}\frac{(x'-w')\cdot(y-x)}{(x'-w')^2(y-x)^2}\Big[\frac{1}{(x-z)^2}-\frac{(x-z)\cdot(y-z)}{2(x-z)^2(y-z)^2}\Big]\mathcal{C}_2-2\,e^{-ik\cdot(w'-z)}\,K_S\,\mathcal{C}_z^{(b)}\Bigg\}, \quad (10.3)
\end{aligned}$$

where $\log\frac{k^+-\Lambda}{\Lambda}\to\log\frac{k^+}{\Lambda}=\log\frac{\vee}{\Lambda}-\log\frac{\vee}{k^+}$. The first two terms are exactly your two log terms, with $(x-z)\cdot(z-x)/[(x-z)^2(z-x)^2]=-1/(x-z)^2$ used.

11 Checks

1. **Nothing is counted twice.** $N_{2b\,\mathrm{UV}}$ is $+\int S$ and the finite and log parts contain $-\int S$: the subtraction line of (10.2) plus the $-2\log$ term of (10.3). Their sum is the original (1.1).

2. $N_{2\,\mathrm{finite}}$ **has no UV log.** At $x \to z$, the main terms of (10.2) give $\int_0^1 d\xi \left[\xi(1-\xi) + [\frac{1}{1-\xi}]_+ + [\frac{1}{\xi}]_+ - 2\right] K_S \mathcal{C}_z^{(b)}$, which the subtraction line cancels term by term at each p^+. The $\mathcal{C}^{(a)}$ part is finite by colour (Step 3).

3. $N_{2\,\log}$ **has no UV log.** At $x \to z$ the three terms of (10.3) go as $(+1+1-2)\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(x-z)^2}\mathcal{C}_z^{(b)} = 0$, and the $\mathcal{C}^{(a)}$ part vanishes by colour.

4. **The only** $\log\mu$ **is in** $N_{2b\,\mathrm{UV}}$**.** It is $\left[2\log\frac{k^+}{\Lambda} - \frac{11}{6}\right] \times \left[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-z)^2\mu_{\overline{MS}}^2}{4}\right]$. If the x integral of the $-2\log$ term in (10.3) is undone, it is $-2\log\frac{k^+}{\Lambda}\,K_S\mathcal{C}_z^{(b)}$, which cancels the $+2\log\frac{k^+}{\Lambda}$ of $N_{2b\,\mathrm{UV}}$ (the $\pm 2N_c$ blocks in the JIMWLK file). What remains of the UV part is $-\frac{11}{6} \times \left[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-z)^2\mu_{\overline{MS}}^2}{4}\right]$, which goes to the Divergent file and combines with Row II.

5. **Large** x **in the log terms.** At $p^+ = 0$ the denominator no longer grows with x, and the first term of (10.3) falls only like $1/(2x^2)$. This is large-distance (IR) behaviour, not UV, and has to be checked in the sum of all $\log(k^+/\Lambda)$ terms in the JIMWLK file.